

Zhao Qunshan

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EDUCATION

- 2013-2017 **Ph.D.**, Geography, Arizona State University, Tempe, AZ, USA
(American Association of Geographers “**J. Warren Nystrom**” Award)
- 2013-2015 **M.A.**, Geography, Arizona State University, School of Geographical Sciences and Urban Planning, Tempe, AZ, USA
- 2009-2013 **B.E.**, Remote Sensing Science and Technology, Wuhan University, Wuhan, China

APPOINTMENTS

- 2024-pres. **Professor in Urban Analytics**, Urban Big Data Centre (UBDC), Urban Studies & Social Policy, School of Social and Political Sciences, University of Glasgow, UK
- 2025-pres. **Deputy Director**, Urban Big Data Centre (UBDC), University of Glasgow, UK
- 2020-2025 **Associate Director**, Urban Big Data Centre (UBDC), University of Glasgow, UK
- 2019-2024 Lecturer (2019-22) and Senior Lecturer (2022-24) in Urban Analytics, Urban Big Data Centre, Urban Studies, School of Social and Political Sciences, University of Glasgow, UK
- 2017-2019 Postdoctoral Research Associate, Knowledge Exchange for Resilience (KER) and Spatial Analysis Research Center (SPARC), Arizona State University, USA

RESEARCH INTERESTS

Empirical: Health Equity and the Societal Determinants of Health, Climate Change Adaptation and Urban Heat, Healthy Housing and Indoor Environments, Building Energy Efficiency and Fuel Poverty, Just Net-Zero Transition, Human Mobility and Activity Space, Urban Green Infrastructure, Urban Resilience, Strategic Foresight for Healthy and Equitable Future Cities.

Methodology: GIScience (Geographic Information Systems, Remote Sensing, Spatial Analysis), Machine Learning/Statistics/Deep Learning/Foundational Models, Operations Research, Big Data Analytics, Urban Climate Modeling and Instrumentation, Low-cost Sensor Network Development.

SELECTED FUNDED RESEARCH

- 2026-2027 **From Smart Data to Health Impacts: Thermal Simulation of Indoor Environments in UK Cold Homes** (Fellowship Mentor. Fellow: Dr Mingyu Zhu. ESRC SDR UK Fellowship, £199,799)
- 2019-2026 **Urban Big Data Centre Data Service Extension (World Class Labs Infrastructure)** (Co-PI - ESRC, £6.7m)
- 2022-2026 **IDEAMAPS: A participatory data-modeling ecosystem for deprived area map production in LMIC cities** (Co-PI - Bill & Melinda Gates Foundation, \$1,690,337)
- 2025-2026 **Sensing Environmental Changes at the fringe of the new Glasgow LEZ: Heat and Air Quality conditions in the Gorbals** (PI - GALLANT Innovation Fund and Crossrod Youth & Community Association, £21,539)
- 2023-2026 **Estimating building energy efficiency and carbon footprint for achieving a net-zero carbon target via urban sensing and AI** (PI – Royal Society, £12,000)
- 2024-2025 **SpaceEPC: Earth Observation Thermal Satellite Data for Measuring Thermal Efficiency of Buildings**, ESA Open Call on Business Application. (Academic PI and led by the Satellite VU Ltd. €489,707, €146,022 (UoG))
- 2023-2025 **The housing environment and energy panel: monitoring indoor residential**

- environments and energy use via low-cost housing sensors** (PI, the Understanding Society Innovation Panel)
- 2023-2025 **Developing and exploring methods to understand human-nature interactions in urban areas using new forms of big data** (Co-PI - ESRC, £292,306)
- 2023 **Sensor-enhanced housing survey for urban heat investigation in London** (PI – the Bureau of Investigative Journalism, £55,470)
- 2021 **Using Mobile Phone App Data to Understand Urban Mobility in Glasgow** (Co-PI – Glasgow City Council, £147,819)

PROFESSIONAL SERVICES

- 2026-pres. Board of Director, the International Association of Chinese Professionals in Geographic Information Sciences (<https://www.cpgis.org/>).
- 2023-pres. MSc Programme External Examiner, Urban Informatics, King's College London.
- 2021-pres. Series Advisory Editors, the Urban Book Series, Springer.
- 2019-pres. Grant Proposal Reviewers, ESRC, EPSRC, Future Leaders Fellowships and US NSF.
- 2025 Grant Assessment Panel Member, ESRC SDR UK Fellowship and UK Space Agency.
- 2022-2024 Grant Assessment Panel Member, UKRI ESRC-Secondary Data Analysis.
- 2022-2024 Member, ESRC Future Data Services Programme Board.
- 2021-2025 Secretary, RGS-IBG Quantitative Methods Research Group (<https://qmrq.github.io/>).
- 2018-2021 Board Member, Spatial Analysis and Modeling Specialty Group, American Association of Geographers (<https://aag-sam.github.io/history.html>).

RESEARCH SUPERVISION & MENTORING

Supervised 8 PhD students (6 completed; graduates now in academic and research posts at Leeds, Edinburgh, Derby, Beijing, and Shanghai) and 8 postdoctoral researchers. Currently mentor on the ESRC Smart Data Research UK Fellowship (thermal environments and health in UK cold homes) and an Understanding Society Fellowship (hidden household underheating). Secured competitive international PhD studentship funding (Glasgow-CSC, SGSSS-Steer and Collaborative, and University-level jointly funded studentships) totalling over £700k.

PUBLICATIONS

57 journal articles, 3 book chapters, 30 conference proceedings, including (Corresponding author):*

- Sun, J., Cao, Y., Grépin, K. A., Li, T., Chen, H., **Zhao, Q.**, & Chi, P. C. (2026). Geospatial analysis of COVID-19 vaccine access in Kenya: The interplay of travel time, perceived availability, and vaccine uptake. *Journal of Global Health*, 16, 04143. <https://doi.org/10.7189/jogh.16.04143>
- Sun, M., Hou, C., Li, Q., Zhang, F., Bardhan, R., & **Zhao, Q.*** (2026). Deciphering exterior: Building energy efficiency prediction with emerging urban big data. *Npj Urban Sustainability*. <https://doi.org/10.1038/s42949-026-00348-7>.
- Jia, S., Yoo, C., Ou, Y., Chen, G., & **Zhao, Q.** (2026). From heat maps to cooling actions: AI-driven citywide hourly mapping of thermal stress, drivers, and targeted cooling. *Sustainable Cities and Society*, 141, 107275. <https://doi.org/10.1016/j.scs.2026.107275>
- Tian, Y., **Zhao, Q.**, Zheng, Z., & Wang, P. (2026). Evaluating urban ventilation corridor through mobile air quality monitoring and environmental justice. *Applied Geography*, 190, 103987. <https://doi.org/10.1016/j.apgeog.2026.103987>
- Feng, Z., **Zhao, Q.***, & Heppenstall, A. (2026). Using agent-based models to simulate the electric vehicle driving behaviours in Great Britain. *Cities*, 172, 106864. <https://doi.org/10.1016/j.cities.2026.106864>
- Feng, Z., **Zhao, Q.***, & Heppenstall, A. (2026). Exploring the influencing factors of public electric vehicle charger usage in Great Britain. *Journal of Transport Geography*, 134, 104667. <https://doi.org/10.1016/j.jtrangeo.2026.104667>
- Li, Y., **Zhao, Q.***, & Wang, M. (2026). Zero-shot traffic flow prediction with foundation models: A comparison with deep learning approaches. *Scientific Reports*. <https://doi.org/10.1038/s41598-026->

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- Ou, Y., Bailey, N., McArthur, D. P., & **Zhao, Q.** (2026). Willingness-to-pay for energy efficient homes: New insights from the 2022 energy crisis in the United Kingdom. *Energy Research & Social Science*, 135, 104700. <https://doi.org/10.1016/j.erss.2026.104700>
- Sun, M., Pan, J., **Zhao, Q.**, & Bardhan, R. (2025). Heat stress dichotomy: Long-term adaptation and acute shock in London domestic environments. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 383(2308), 20240567. <https://doi.org/10.1098/rsta.2024.0567>
- Cao, Y., Li, T., Chen, H., **Zhao, Q.**, Sun, J., Grépin, K. A., & Kang, J.-Y. (2025). Evaluating geographic accessibility to COVID-19 vaccination across 54 countries/regions. *BMJ Global Health*, 10(2). <https://doi.org/10.1136/bmjgh-2024-017761>
- Ou, Y., Bailey, N., McArthur, D. P., & **Zhao, Q.** (2025). The price premium of residential energy performance certificates: A scoping review of the European literature. *Energy and Buildings*, 332, 115377. <https://doi.org/10.1016/j.enbuild.2025.115377>
- Li, Y., **Zhao, Q.***, & Wang, M. (2025). High-resolution traffic flow data from the urban traffic control system in Glasgow. *Scientific Data*, 12(1), 253. <https://doi.org/10.1038/s41597-025-04494-y>
- Li, Y., **Zhao, Q.***, & Wang, M. (2024). Understanding urban traffic flows in response to COVID-19 pandemic with emerging urban big data in Glasgow. *Cities*, 154, 105381. <https://doi.org/10.1016/j.cities.2024.105381>
- Shin, H., Silverman, E., Li, Y., Tian, Y., & **Zhao, Q.** (2024). Did the Implementation of Low Emission Zone in Glasgow Change the Traffic Flow and Air Quality? *Findings*. <https://doi.org/10.32866/001c.123382>
- Chen, H., Cao, Y., Feng, L., **Zhao, Q.**, & Torres, J. R. V. (2023). Understanding the spatial heterogeneity of COVID-19 vaccination uptake in England. *BMC Public Health*, 23(1), Article 1. <https://doi.org/10.1186/s12889-023-15801-w>
- Park, Y., **Zhao, Q.***, Guldmann, J., & Wentz, E. (2023). Quantifying the Cumulative Cooling Effects of 3D Building and Tree Shades with High Resolution Thermal Imagery in a Hot Arid Urban Climate.” *Landscape and Urban Planning*, 240, 104874. <https://doi.org/10.1016/j.landurbplan.2023.104874>
- Hao, T., **Zhao, Q.***, & Huang, J. (2023). Optimization of tree locations to reduce human heat stress in an urban park.” *Urban Forestry & Urban Greening*, 86, 128017. <https://doi.org/10.1016/j.ufug.2023.128017>
- Sun, M., Han, C., Nie, Q., Xu, J., Zhang, F., & **Zhao, Q.*** (2022). Understanding building energy efficiency with administrative and emerging urban big data by deep learning in Glasgow. *Energy and Buildings*, 273, 112331. <https://doi.org/10.1016/j.enbuild.2022.112331>
- Zhao, Q.***, Li, Z., Shah, D., Fischer, H., Solis, P., & Wentz, E. A. (2021) Understanding the interaction between human activities and physical health under extreme heat environment in Phoenix, Arizona. *Health and Place*, 102691. <https://doi.org/10.1016/j.healthplace.2021.102691>
- Wu, Y., Zhan, Q., & **Zhao, Q.** (2020). Long-term air pollution exposure impact on COVID-19 morbidity in China. *Aerosol and Air Quality Research*, 20. <https://doi.org/10.4209/aaqr.2020.07.0413>

HONOURS & AWARDS

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| 2024 | GIS Cup 2024 Award (3rd place), 32nd ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (with PhD student Zixin Feng). |
| 2024 | Finalist, Early Career Researcher Award, RTPI Awards for Research Excellence (with HKU PhD student Tongping Hao). |
| 2022 | Shortlist, Most Innovative Transport Project of the Year, Scottish Transport Awards (Glasgow City Council & UBDC). |
| 2021 | Fellowship of Recognising Excellence in Teaching (RET), University of Glasgow. |
| 2020 | Global Young Scientist Award, World Geospatial Developers Conference. |
| 2018 | J. Warren Nystrom Award (outstanding paper from a recent dissertation), American Association of Geographers. |